# Introduction to Robotics, Homework 1

The AR-60TM arm

Forward and inverse kinematics, differential kinematics, singularities

MIPT

Center for Cognitive Modeling

# What this homework is

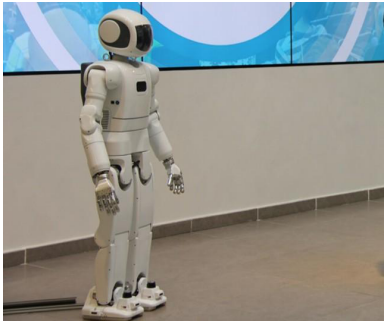
*One arm, six tasks, eighteen hours*

AR-601M is a 41-DOF humanoid robot. This homework treats one of its two arms in isolation: a 6-DOF serial chain, shoulder to tool, with a spherical wrist.

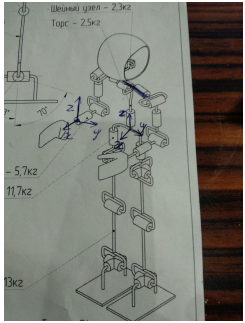
- **Forward** kinematics: joint angles → tool pose.
- **Inverse** kinematics, analytical and numerical: tool pose → joint angles.
- **Differential kinematics** and **singularities**.

About 18 hours of work covering weeks 1–3. This deck is a pointer; the full statement, with every derivation, requirement and hint, is the handout linked on the last frame.

# The mechanism: AR-601M



(a) The AR-601M arm.




(b) The blueprint, annotated in blue.

Figure: The robot, and the blueprint it is modelled from; lengths are not legible on the blueprint, see handout §2.

# The six tasks

- **1.** Kinematic model
- **2.** FK: angles → arm and pose
- **3.** Analytical IK: pose → all feasible branches
- **4.** Numerical IK: solve from multiple seeds
- **5.** Differential kinematics
- **6.** Singularity and solution-count maps

Full requirements and evidence for every task are in handout §4.

# Required visualizations

1. **FK:** enter six joint angles; draw the manipulator and print the resulting homogeneous matrix $T$.
2. **Analytical IK:** enter a pose from FK or a position and ZYX Euler orientation; draw *every distinct feasible* configuration and print each complete six-angle vector.
3. **Singularities:** plot singularity information and a fixed-orientation workspace slice. Use a continuous color scale for the number of feasible IK configurations, from zero to the maximum attained in this case.

Mark unresolved targets and continuous families separately from finite counts.

# What you hand in

## The packaging contract

- A `pixi.toml` or a `Dockerfile` at the repository root.
- One command reproduces the report's numbers, example images and maps from a clean checkout.
- Three operations: `FK(q, robot)`, `IK(T, robot)`, `DifferentialKinematics(q, robot)` — any language.

Packaging is what is graded, not the language used to write it.

# The full statement

Everything above is a pointer. The handout has every derivation, every requirement and every hint.

Read handout §3 and §4 before writing any code.

hw01_handout.pdf on GitHub



**The handout**

# Deserve “A” grade!

– Oleg Bulichev

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